Milestone Two: Cleaning, Testing, and Exploring Socioeconomic Well-Being in BC

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```
#bc_sei_data_raw <- bcdc_get_data("b-c-socioeconomic-index")
```

```
#write_csv(bc_sei_data_raw, "bc_sei_data_raw.csv")
```

After writing the dataset onto my computer, I have made a copy of the dataset to keep the original file.

```
socioeconomic_index_raw <- read_csv("bc_sei_data_raw.csv")
```

```
Rows: 319 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (1): MUN_NAME_2021
dbl (10): CSD_UID, CALENDAR_YEAR, AVERAGE_AGE, POPULATION_ESTIMATE, TOTAL_IN...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Cleaning

Let's get a sense of the data here.

```
head(socioeconomic_index_raw)
```

```
# A tibble: 6 x 11
  CSD_UID MUN_NAME_2021   CALENDAR_YEAR AVERAGE_AGE POPULATION_ESTIMATE
    <dbl> <chr>                   <dbl>       <dbl>               <dbl>
```

```
1 5901003 Elkford                  2023        37.3                2949
2 5901006 Sparwood                 2023        38.2                4452
3 5901012 Fernie                   2023        38.6                6778
4 5901017 East Kootenay A          2023        45.8                2003
5 5901019 East Kootenay B          2023        45.8                2148
6 5901022 Cranbrook                2023        43.5               22057
# i 6 more variables: TOTAL_INDEX_0_100 <dbl>, SEI_INDEX_0_100 <dbl>,
#   ECON_0_100 <dbl>, EDUC_0_100 <dbl>, HEALTH_0_100 <dbl>,
#   COMMUNITY_0_100 <dbl>
```

First, I will get rid of columns that I do not need.

```
cleaned_socioeconomic_index <- socioeconomic_index_raw |>
  select(-CALENDAR_YEAR)
```

Now, I will only include the municipalities of interest.

```
cleaned_socioeconomic_index <- cleaned_socioeconomic_index |>
  filter(
    MUN_NAME_2021 %in% c(
    "Anmore", "Belcarra", "Bowen Island", "Burnaby",
    "Coquitlam", "Delta", "Langley", "Lions Bay", "Maple Ridge",
    "Metro Vancouver A", "New Westminster", "North Vancouver",
    "Pitt Meadows", "Port Coquitlam", "Port Moody", "Richmond",
    "Surrey", "Vancouver", "West Vancouver", "White Rock", "Central Saanich",
    "Colwood", "Esquimalt", "Highlands", "Juan de Fuca (Part 1)",
    "Juan de Fuca (Part 2)", "Langford", "Metchosin", "North Saanich",
    "Oak Bay", "Saanich", "Sidney", "Sooke", "Victoria", "View Royal"
  )
)
```

In the dataset, some municipalities are named twice because they contain sub-regions or districts. It is important that we give these better identifiers so that we can be more precise in our analysis.

```
cleaned_socioeconomic_index <- cleaned_socioeconomic_index |>
  mutate(
    MUN_NAME_2021 = case_when(
    CSD_UID == "5915002" ~ "City of Langley",
    CSD_UID == "5915001" ~ "Township of Langley",
    CSD_UID == "5915020" ~ "Electoral Area A (UBC)",
```

```
    CSD_UID == "5915051" ~ "City of North Vancouver",
    CSD_UID == "5915046" ~ "District of North Vancouver",
    CSD_UID == "5917054" ~ "Juan de Fuca (Part 1, Shirley, Jordan River, Malahat)",
    CSD_UID == "5917056" ~ "Juan de Fuca (Part 2, Port Renfrew)",
    TRUE ~ MUN_NAME_2021
  )
)
```

Let's also rename the columns

```
cleaned_socioeconomic_index <- cleaned_socioeconomic_index |>
  rename(
    Municipality = MUN_NAME_2021,
    Average_Age = AVERAGE_AGE,
    Population_2023 = POPULATION_ESTIMATE,
    Total_Index = TOTAL_INDEX_0_100,
    Socioeconomic_Index = SEI_INDEX_0_100,
    Economics = ECON_0_100,
    Education = EDUC_0_100,
    Health = HEALTH_0_100,
    Community = COMMUNITY_0_100
  )
```

Now, lets make it so that the index data only goes up to two decminal points.

```
index_decimal <- 2

cleaned_socioeconomic_index <- cleaned_socioeconomic_index |>
  mutate(
    across(
      c(
        Total_Index,
        Socioeconomic_Index,
        Economics,
        Education,
        Health,
        Community
    ),
    round, digits = index_decimal
  )
)
```

```
cleaned_socioeconomic_index |>
  as_tibble()
```

```
# A tibble: 37 x 10
   CSD_UID Municipality           Average_Age Population_2023 Total_Index
     <dbl> <chr>                        <dbl>           <dbl>       <dbl>
 1 5915001 Township of Langley           39.7          154103        56.1
 2 5915002 City of Langley               41.4           33626        46.8
 3 5915004 Surrey                        38.4          653040        52.0
 4 5915007 White Rock                    52.1           24294        51.0
 5 5915011 Delta                         41.8          120048        54.1
 6 5915015 Richmond                      42.1          235082        54.9
 7 5915020 Electoral Area A (UBC)        37.8           21150        60.0
 8 5915022 Vancouver                     40.9          732593        55.8
 9 5915025 Burnaby                       40.7          283588        56.0
10 5915029 New Westminster               40.7           88410        54.2
# i 27 more rows
# i 5 more variables: Socioeconomic_Index <dbl>, Economics <dbl>,
#   Education <dbl>, Health <dbl>, Community <dbl>
```

```
age_decimal <- 0

cleaned_socioeconomic_index <- cleaned_socioeconomic_index |>
  mutate(
    across(
      c(
        Average_Age
      ),
      round, digits = age_decimal
  )
)

cleaned_socioeconomic_index |>
  as_tibble()
```

```
# A tibble: 37 x 10
   CSD_UID Municipality           Average_Age Population_2023 Total_Index
     <dbl> <chr>                        <dbl>           <dbl>       <dbl>
 1 5915001 Township of Langley             40          154103        56.1
 2 5915002 City of Langley                 41           33626        46.8
 3 5915004 Surrey                          38          653040        52.0
```

```
 4 5915007 White Rock                      52          24294        51.0
 5 5915011 Delta                           42         120048        54.1
 6 5915015 Richmond                        42         235082        54.9
 7 5915020 Electoral Area A (UBC)          38          21150        60.0
 8 5915022 Vancouver                       41         732593        55.8
 9 5915025 Burnaby                         41         283588        56.0
10 5915029 New Westminster                 41          88410        54.2
# i 27 more rows
# i 5 more variables: Socioeconomic_Index <dbl>, Economics <dbl>,
#   Education <dbl>, Health <dbl>, Community <dbl>
```

Tests

I have two basic tests I want to run. The first is to make sure all variables are proper class. I am doing this by using the stopifnot function, which will stop running the code if it detects a variable that is not the proper class. The second is to make sure there are no negative values in the numeric variables. This code uses a logical if/else function to print out either 'passed' or 'stop' statements depending on whether number less than zero are found in each variable.

```
stopifnot(
  "Municipality must be a character" = is.character(cleaned_socioeconomic_index$Municipality)
  "Population must be numeric" = is.numeric(cleaned_socioeconomic_index$Population_2023),
  "Average age must be numeric" = is.numeric(cleaned_socioeconomic_index$Average_Age),
  "Total index must be numeric" = is.numeric(cleaned_socioeconomic_index$Total_Index),
  "SEI Index muste be numeric" = is.numeric(cleaned_socioeconomic_index$Socioeconomic_Index)
  "Community must be numeric" = is.numeric(cleaned_socioeconomic_index$Community),
  "Health must be numeric" = is.numeric(cleaned_socioeconomic_index$Health),
  "Education must be numeric" = is.numeric(cleaned_socioeconomic_index$Education),
  "Economics must be numeric" = is.numeric(cleaned_socioeconomic_index$Economics)
)
```

```
numeric_variables <- c(
  "Population_2023", "Average_Age", "Total_Index", "Socioeconomic_Index",
  "Community", "Health", "Education", "Economics"
)

for(
  SEI_var in numeric_variables) {
    if(any(cleaned_socioeconomic_index[[SEI_var]] < 0, na.rm = TRUE)) {
      stop(
        paste("Stop, there are negative values in the", SEI_var, "variable"))
```

```
 } else {
      message(
        paste("Passed, there are no negative values in the", SEI_var, "variable")
  )
 }
}
```

```
Passed, there are no negative values in the Population_2023 variable
```

```
Passed, there are no negative values in the Average_Age variable
```

```
Passed, there are no negative values in the Total_Index variable
```

```
Passed, there are no negative values in the Socioeconomic_Index variable
```

```
Passed, there are no negative values in the Community variable
```

```
Passed, there are no negative values in the Health variable
```

```
Passed, there are no negative values in the Education variable
```

```
Passed, there are no negative values in the Economics variable
```

Looks like all the tests passed and there are no negative numbers where there should not be.

Exploratory Data Analysis

Before making any graphs, I am gonna get a glimpe of the data and run some basic descriptive statistics to get a sense of central tendency and variability in the data.

```
head(cleaned_socioeconomic_index)
```

```
# A tibble: 6 x 10
  CSD_UID Municipality        Average_Age Population_2023 Total_Index
    <dbl> <chr>                     <dbl>           <dbl>       <dbl>
1 5915001 Township of Langley          40          154103        56.1
2 5915002 City of Langley              41           33626        46.8
3 5915004 Surrey                       38          653040        52.0
4 5915007 White Rock                   52           24294        51.0
5 5915011 Delta                        42          120048        54.1
6 5915015 Richmond                     42          235082        54.9
# i 5 more variables: Socioeconomic_Index <dbl>, Economics <dbl>,
#   Education <dbl>, Health <dbl>, Community <dbl>
```

```
tail(cleaned_socioeconomic_index)
```

```
# A tibble: 6 x 10
  CSD_UID Municipality                    Average_Age Population_2023 Total_Index
    <dbl> <chr>                                 <dbl>           <dbl>       <dbl>
1 5917044 Langford                                 38           55064        54.2
2 5917047 View Royal                               43           12578        54.4
3 5917049 Highlands                                43            2896        58
4 5917052 Sooke                                    43           16796        47.7
5 5917054 Juan de Fuca (Part 1, Shirley~           50            5527        48.6
6 5917056 Juan de Fuca (Part 2, Port Re~           50             430        47.1
# i 5 more variables: Socioeconomic_Index <dbl>, Economics <dbl>,
#   Education <dbl>, Health <dbl>, Community <dbl>
```

```
glimpse(cleaned_socioeconomic_index)
```

```
Rows: 37
Columns: 10
$ CSD_UID             <dbl> 5915001, 5915002, 5915004, 5915007, 5915011, 59150~
$ Municipality        <chr> "Township of Langley", "City of Langley", "Surrey"~
$ Average_Age         <dbl> 40, 41, 38, 52, 42, 42, 38, 41, 41, 41, 40, 49, 40~
$ Population_2023     <dbl> 154103, 33626, 653040, 24294, 120048, 235082, 2115~
$ Total_Index         <dbl> 56.12, 46.79, 52.03, 50.95, 54.09, 54.87, 60.01, 5~
$ Socioeconomic_Index <dbl> 56.71, 46.49, 51.50, 53.01, 54.79, 54.38, 57.09, 5~
$ Economics           <dbl> 58.70, 51.75, 55.36, 51.16, 56.43, 48.57, 45.04, 5~
$ Education           <dbl> 52.75, 44.82, 46.15, 53.31, 50.12, 57.94, 66.64, 5~
$ Health              <dbl> 51.84, 49.06, 52.46, 44.77, 50.20, 54.29, 64.36, 5~
$ Community           <dbl> 54.85, 44.70, 52.32, 52.88, 55.22, 54.18, 55.48, 5~
```

```
summary(cleaned_socioeconomic_index)
```

```
    CSD_UID        Municipality         Average_Age    Population_2023
 Min.   :5915001   Length:37          Min.   :38.00   Min.   :   430
 1st Qu.:5915029   Class :character   1st Qu.:41.00   1st Qu.: 13162
 Median :5915062   Mode  :character   Median :42.00   Median : 24294
 Mean   :5915846                      Mean   :43.57   Mean   : 91549
 3rd Qu.:5917034                      3rd Qu.:46.00   3rd Qu.:100497
 Max.   :5917056                      Max.   :55.00   Max.   :732593
  Total_Index    Socioeconomic_Index   Economics       Education
 Min.   :46.79   Min.   :43.54       Min.   :33.72   Min.   :41.43
 1st Qu.:51.16   1st Qu.:52.29       1st Qu.:53.02   1st Qu.:49.59
 Median :54.87   Median :55.19       Median :55.66   Median :52.98
 Mean   :55.12   Mean   :55.39       Mean   :55.53   Mean   :53.56
 3rd Qu.:58.40   3rd Qu.:57.84       3rd Qu.:58.70   3rd Qu.:57.06
 Max.   :65.62   Max.   :66.46       Max.   :65.62   Max.   :70.13
     Health        Community
 Min.   :40.52   Min.   :44.70
 1st Qu.:50.20   1st Qu.:52.42
 Median :51.84   Median :53.91
 Mean   :52.19   Mean   :54.01
 3rd Qu.:54.86   3rd Qu.:56.54
 Max.   :64.36   Max.   :60.47
```

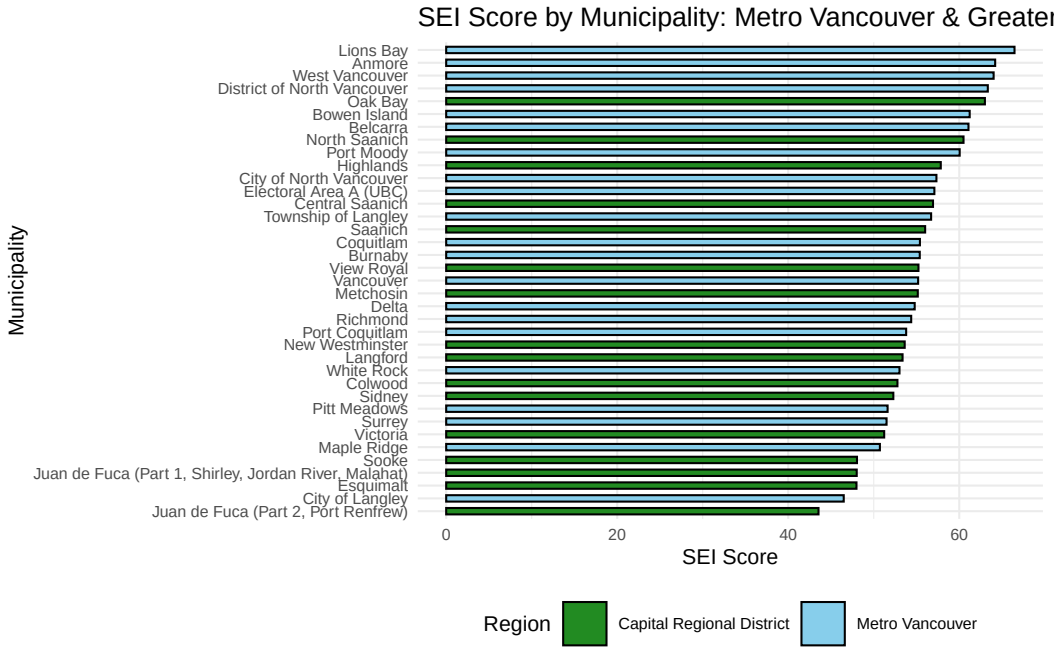
Bar Graphs

First, I want to create a bar graph that shows SEI score by each municipality. I also reordered the municipalities so that they are ordered from highest SEI score to lowest SEI score. I create a new variable that encapsulates all the municipalities in Metro Vancouver. Then, I use a logical if_else function to mutate the dataset so that the rest of the municipalities not included in Metro Vancouver are placed in the CRD.

```
metro_vancouver <- c(
  "Anmore", "Belcarra", "Bowen Island", "Burnaby", "City of Langley",
  "City of North Vancouver", "Coquitlam", "Delta", "District of North Vancouver",
  "Electoral Area A (UBC)", "Lions Bay", "Maple Ridge", "New Westminister",
  "Pitt Meadows", "Port Coquitlam", "Port Moody", "Richmond", "Surrey",
  "Township of Langley", "Vancouver", "West Vancouver", "White Rock"
)

cleaned_socioeconomic_index |>
```

```
  mutate(
    Region = if_else(
      Municipality %in% metro_vancouver, "Metro Vancouver", "Capital Regional District"
      ),
    Municipality = reorder(Municipality, Socioeconomic_Index)
      ) |>
  ggplot(
    aes(
      x = Municipality, y = Socioeconomic_Index, fill = Region
      )
    ) +
      geom_col(
        colour = "black", width = 0.5
    ) +
      coord_flip() +
      scale_fill_manual(
        values = c("Metro Vancouver" = "skyblue", "Capital Regional District" =
        "forestgreen")
      ) +
      labs(
        title = "SEI Score by Municipality: Metro Vancouver & Greater Victoria",
        x = "Municipality",
        y = "SEI Score",
        fill = "Region"
    ) +
      theme_minimal(
        base_size = 8
    ) +
      theme(
        legend.position = "bottom"
      )
```

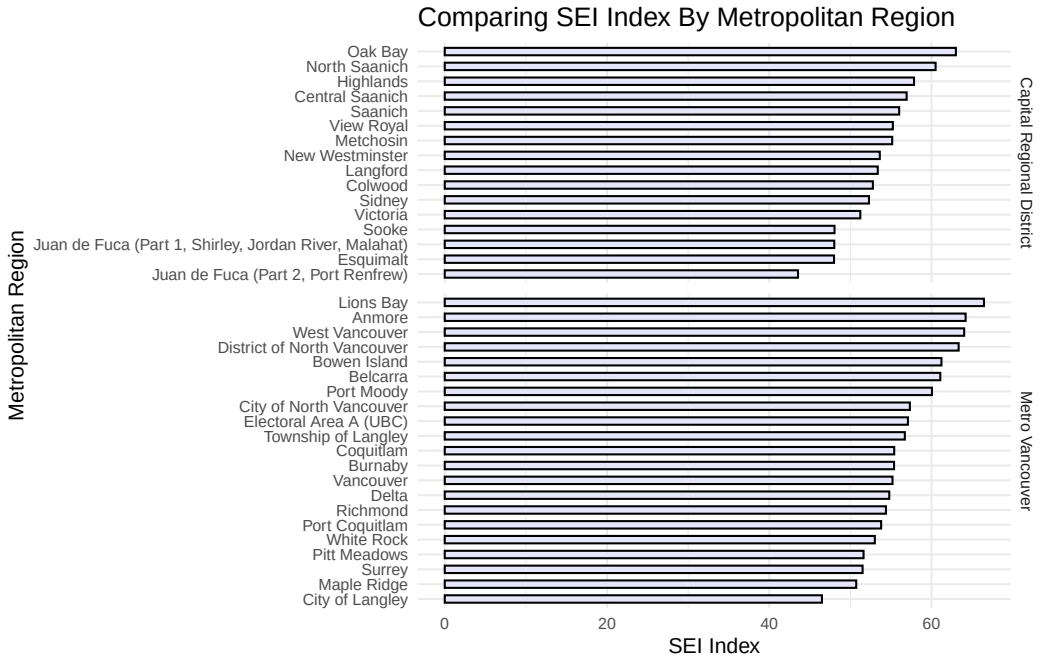


Alternatively, here is a bar graph that separates SEI scores by metropolitan region. This gives us an interesting peek into well-being index scores by metropolitan region.

```
metro_vancouver <- c(
  "Anmore", "Belcarra", "Bowen Island", "Burnaby", "City of Langley",
  "City of North Vancouver", "Coquitlam", "Delta", "District of North Vancouver",
  "Electoral Area A (UBC)", "Lions Bay", "Maple Ridge", "New Westminister",
  "Pitt Meadows", "Port Coquitlam", "Port Moody", "Richmond", "Surrey",
  "Township of Langley", "Vancouver", "West Vancouver", "White Rock"
)

cleaned_socioeconomic_index |>
  mutate(
    Region = if_else(
      Municipality %in% metro_vancouver, "Metro Vancouver", "Capital Regional District"
      ),
    Municipality = reorder(Municipality, Socioeconomic_Index)
      ) |>
    ggplot(
      aes(
        x = Municipality, y = Socioeconomic_Index
        )
      ) +
```

```
  geom_col(
    fill = "lavender", colour = "black", width = 0.5
  ) +
  coord_flip(
  ) +
  facet_grid(
    Region ~ ., scales = "free_y", space = "free"
  ) +
  labs(
    title = "Comparing SEI Index By Metropolitan Region",
    x = "Metropolitan Region",
    y = "SEI Index"
  ) +
  theme_minimal(
    base_size = 8
  )
```



Based on this graph, we can see that SEI index scores are relatively higher in the Metro Vancouver region than the CRD.

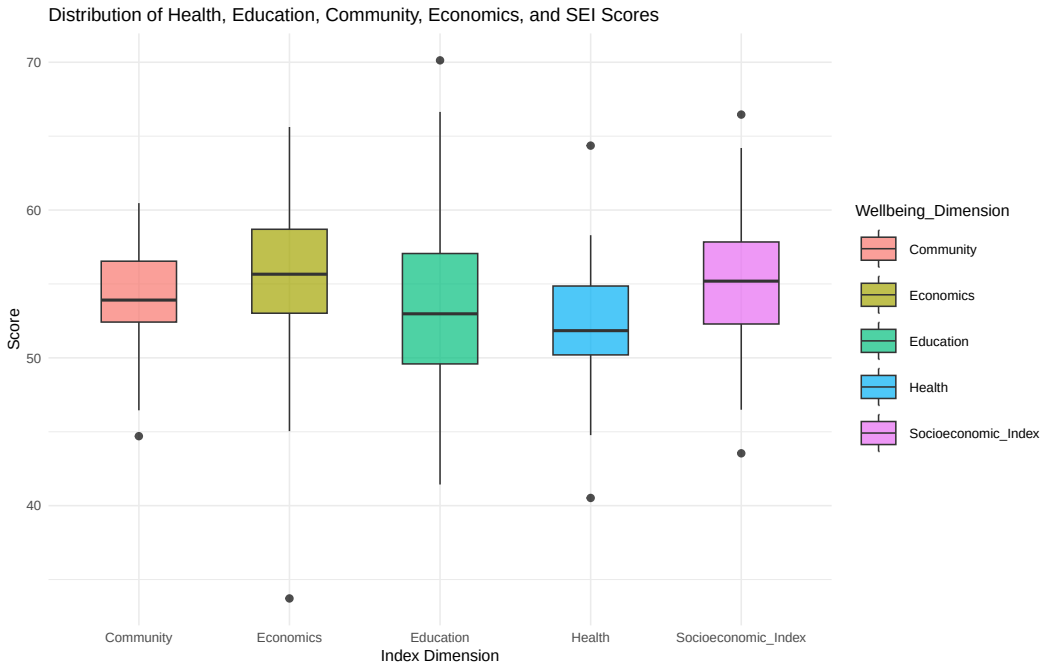
Box Plot

Lastly, we can create a box plot to see the distribution of scores across key numeric variables in my dataset. We will look at education, heath, community, economics, and SEI index.

First, I am going to use the pivot_longer function to reshape my data, the names_to argument makes a new column at the end of the dataset and names each dimension (Education, Heath, etc...) in this new column for each municipality. The values_to argument prints the scores to each respective dimension into a new column. The reason I am reshaping the data is because it makes it easier for ggplot to differentiate each dimension. It creates five separate rows for each municipality, instead of each municiaplity only having one row containing all five dimensions.

```
cleaned_socioeconomic_index |>
  pivot_longer(
    c(
      Education, Health, Community, Economics, Socioeconomic_Index
      ),
    names_to = "Wellbeing_Dimension",
    values_to = "Score"
    ) |>

ggplot(
  aes(
    x = Wellbeing_Dimension, y = Score, fill = Wellbeing_Dimension
    )
  ) +
  geom_boxplot(
    alpha = 0.7, width = 0.5, outlier.color = "black"
  ) +
  labs(
    title = "Distribution of Health, Education, Community, Economics, and SEI Scores",
    x = "Index Dimension",
    y = "Score"
  ) +
  theme_minimal(
    base_size = 6
  )
```



Based on this graph, we can see that economics has the highest median index score, while health has the lowest. The median SEI index is floating around a score of 55 out of 100. It also seems that education has the largest variability within the 25th and 75th percentile and community has the lowest variability within the 25th and 75th percentile.